

Meng Wang

POSTDOCTORAL FELLOW · THE UNIVERSITY OF TEXAS AT AUSTIN
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Education

University of Houston

Houston, TX

PHD, ENVIRONMENTAL ENGINEERING | GPA: 3.8/4.0

2018 - 2023

- Dissertation: "Mechanistic Study on Polymer Coatings to Control Mineral Scaling in Desalination Treatment"

China University of Petroleum Beijing

Beijing, China

MS, PETROLEUM ENGINEERING | GPA: 4.0/4.0

2015 - 2018

- Thesis: "Numerical Study of Proppant Transport with Supercritical CO₂ in Fracture"

Northeast Petroleum University

Heilongjiang, China

BE, PETROLEUM ENGINEERING | GPA: 4.39/5.0

2011 - 2015

- Thesis: "Hydraulic Fracturing Model Featuring Initiation Beyond the Wellbore Wall For Directional Well in Coal Bed"

Research Experience

The University of Texas at Austin

08/2023–present

Postdoctoral Fellow

Advisor: Dr. Lynn Katz

- Conducted research in the field of ion pairing and explored the influence of ion pairing and solvation on the sorption and diffusion of ions in functionalized membrane materials.
- Analyzed relationships between ion pairing, polymer functionality, structural characteristics, and system properties including dielectric constants, water content, pH, and electrical double layer characteristics.
- Established comprehensive methods for characterizing solute interactions including molecular dynamic(MD), Raman, and dielectric spectroscopy to determine the ionic association constant in the solution/solvent or membrane
- Investigated variations in ion pairing between bulk solution properties and membranes, and revealed the effects of varied bulk solution/solvent solutions and ion association
- Developed a model to correlate changes in salt concentration within the membrane with those in the external salt solution
- Delivered 2 presentations at ACS Spring 2025 Meeting, and Association of Environmental Engineering and Science Professors (AEESP) 2025 Research and Education Conference, respectively

University of Houston

08/2018–05/2023

Ph.D. Student/Research Assistant

Advisor: Dr. Debora F. Rodrigues

- Synthesized and characterized different kinds of novel and advanced polymer coatings (i.e., Graphene Oxide, Zwitterion amphiphilic copolymers)
- Conducted gypsum scaling on different membrane surfaces using a bench-scale cross-flow reverse osmosis filtration system
- Quantified gypsum nucleation and growth rate at organic-water interfaces and characterized mineral crystal properties with various techniques
- Mentored 2 graduate students and 1 undergraduate student (funded by research experience for undergraduates-REU)
- Published 4 first-author and 3 co-authored papers (i.e., [DOI: 10.1021/acs.est.1c04143](https://doi.org/10.1021/acs.est.1c04143))
- Delivered 2 presentations at ACS Southwest Regional Meeting, and Water Quality Technology Conference 2019, respectively

Argonne National Laboratory

01/2019-11/2022

Visiting Student

- Prepared polymer coatings for X-ray scattering tests including Grazing Incidence Small Angle X-ray Scattering (GISAXS), and Grazing Incidence Wide-Angle X-ray Scattering (GIWAXS) at Beamline 12-ID-B of the Advanced Photon Source
- Observed heterogeneous mineral nucleation and growth kinetics on different polymer coatings in real-time at the early stages of mineral scaling using in-situ GISAXS
- Characterized the crystallinity for mineral particles formed on polymer coatings using ex-situ GIWAXS
- Converted 2-dimensional images to 1-dimensional data using MATLAB GISAXSshop program, and Igor Pro-6.3
- Conducted model fitting of 1D GISAXS line-cut scattering curves using Irena SAS Igor macro and analyzed the evolution of mineral nanoparticles formed on substrates

China University of Petroleum Beijing

Master Student

09/2015-07/2018

Advisor: Dr. Haizhu Wang

- Performed rock-core scale fracturing with supercritical CO₂ (SC-CO₂) fluid
- Simulated proppant transport with SC-CO₂ in fractures using ANSYS FLUENT
- Designed experimental devices for proppant transport with SC-CO₂ in the fracture
- Published 1 first author and 6 co-author papers in various Journals (i.e., [DOI: 10.1002/ghe.1803](https://doi.org/10.1002/ghe.1803))

Northeast Petroleum University

Undergraduate student

02/2014-07/2015

Advisor: Dr. Yuwei Li

- Developed a model to calculate the initiation pressure that occurs along with coal-bed cleats
- Wrote a simulator with over 10k lines of code in C/C++ to calculate the initiation pressure
- Published results in the Journal of Geophysics and Engineering ([DOI:10.1088/1742-2132/13/4/536](https://doi.org/10.1088/1742-2132/13/4/536))

Working Experience

Environmental Engineering

Teaching Assistant

Houston, TX

Aug. 2022 - Nov. 2022

- Collaborated with instructor to grade homework, and exams
- Assisted instructor in answering student questions
- Delivered lectures on homework solutions

Baker Tilly Beijing

Consultant Intern

Beijing, China

May. 2018 - July. 2018

- Provided support to the project manager in conducting internal audits for clients.
- Contributed to the preparation of proposals for client talent promotion evaluations.
- Assisted with project status investigations, questionnaire creation, and report writing.

ConocoPhillips China Incorporation

Research Intern

Beijing, China

Jul. 2017 - Sept. 2017

- Managed oil well production data using PL/SQL, and performed data visualization with Python
- Analyzed drilling daily logs and identified pros and cons of various wellbore structure designs
- Data processing of core sample test and sand size distribution analysis using MATLAB
- Proppant size screening for Peng Lai oilfield based on the sand size distribution analysis

Publications

R Jacovone, **M Wang**, J Soares, S Sakata, D Rodrigues, Investigating the influence of polymaleic acid on reverse osmosis membrane performance to mitigate gypsum scaling, *Desalination* 2026, 619, 119544

M Wang, M Sarma, S. Lounder, A Mondal, L Muthusamy, G Koley, A Asatekin, D Rodrigues, Organic Fouling on Zwitterionic Amphiphilic Copolymers: Implications in Biofouling, *ACS Applied Materials & Interfaces*, 2025, 17, 20, 30149–30160

M Wang, X Zuo, R Jacovone, R O'Hara, A Mondal, A, Asatekin, D Rodrigues, Influence of Zwitterionic Amphiphilic Copolymers on Heterogeneous Gypsum Formation: A Promising Approach for Scaling Resistance, *Water Research*, 2024, 122439

F Chen, Y Wang, R Bi, Y Pan, **M Wang**, Molecular insights into CO₂ enhanced hydrocarbon recovery and its sequestration in multiscale shale reservoirs, *Chemical Engineering Journal*, 2024, 498: 155913

Y Wang, Q Sun, F Chen, **M Wang**, Multiscale Model for Hydrogen Transport and Storage in Shale Reservoirs, *SPE Journal*, 2024, 1-27

M Wang, H Nguyen, S Lounder, A Asatekin, D Rodrigues, Calcium Sulfate Formation on Different Zwitterionic Amphiphilic Copolymer Substrates for Salt Water Treatment, *ACS Applied Polymer Materials*, 2022, 4 (10): 7090-7101

H Arabaghian, **M Wang**, J Ordonez, D Rodrigues, Zwitterionic polymers in biofouling and inorganic fouling mechanisms, *60 Years of the Loeb-Sourirajan Membrane*, 2022, 33-70. **(Book Chapter, Co-first Author)**

M Wang, B Cao, Y Hu, D Rodrigues, Mineral Scaling on Reverse Osmosis Membranes: Role of Mass, Orientation, and Crystallinity on Permeability, *Environmental Science & Technology*, 2021, 55 (23): 16110-16119

A Ansari, J Peña-Bahamonde, **M Wang**, D Shaffer, Y Hu, D Rodrigues, Polyacrylic acid-brushes tethered to graphene oxide coating
Oct. 2025

for simultaneous scaling and biofouling mitigation on reverse osmosis membranes, *Journal of Membrane Science*, 2021, 630, 119308

T Ge, **M Wang**, F Li, F Min, M Zhang. Study and spectroscopy characterization of carbon and oxygen structures of Gaoyang coking coal, *Journal of China Coal Society*, 2021, 46 (3):1024-1031

H Wang, G Li, Z Shen, Z He, Q Liu, B Zhu, Y Wang, **M Wang**, Expulsive force in the development of CO₂ sequestration: application of SC-CO₂ jet in oil and gas extraction, *Frontiers in Energy*, 2019, 13:1-8.

M Wang, H Wang, G Li, B Yang, Q Lu, Numerical Study of Proppant Transport with Supercritical CO₂ in Fracture, *China Petroleum Machinery*, 2018, 46(11): 72-78

H Wang, **M Wang**, B Yang, Q Lu, Y Zheng, H Zhao, Numerical Study of SC-CO₂ and Proppant Transport in Different Geometrical Fractures, *Greenhouse Gases: Science and Technology*, 2018, 8 (5):898-910

H Wang, G Li, Z He, S Tian, **M Wang**, B Yang, Q Lu, L Weng, Experimental Investigation on Abrasive Supercritical CO₂ Jet Perforation, *Journal of CO₂ Utilization*, 2018, 28:59–65

H Wang, G Li, Z He, Z Shen, X Li, Z Zhang, **M Wang**, B Yang, Y Zheng, L Shi, Analysis of rock fracture mechanism caused by supercritical CO₂, *Rock and Soil Mechanics*, 2018, 39, 10: 3589-3596

H Wang, G Li, L Zhao, Y Wang, Y Liu, Q Liu, **M Wang**, F Gao, M Cai, Throttle characteristics of multi-stage circumfluence nozzle during the separate-layer injection of CO₂, *Petroleum*, 2018, 4(2):197-197

Y Wang, J Yuan, Z Wang, Y Yin, **M Wang**, W Wang, Experimental study on the effect of mechanical vibration on cement stone strength, *Natural Gas Exploration and Development*, 2018, 41(1):96-100.

H Wang, G Li, Z He, Z Shen, **M Wang**, Y Wang, Mechanism Study on Rock Breaking with Supercritical Carbon Dioxide Jet, *Atomization and Sprays*, 2017, 27(5):383–394

X Li, G Li, H Wang, S Tian, X Song, P Lu, **M Wang**, A Unified Model for Wellbore Flow and Heat Transfer in Pure CO₂ Injection for Geological Sequestration, EOR and Fracturing Operations, *International Journal of Greenhouse Gas Control*, 2017, 57:102-115

Y Li, D Jia, **M Wang**, J Liu, C Fu, X Yang, C Ai, Hydraulic Fracturing Model Featuring Initiation Beyond the Wellbore Wall For Directional Well in Coal Bed, *Journal of Geophysics and Engineering*, 2016, 13:536–548

X Li, G Li, H Wang, P Lu, S Tian, R Yang, **M Wang**, A Coupled Model for Predicting Flowing Temperature and Pressure Distribution in Drilling Ultra-short Radius Radial Wells, *IADC/SPE Asia Pacific Drilling Technology Conference*, 2016, SPE-180597-MS

Patent

H Wang, G Li, J Li, Z Huang, **M Wang**, B Yang, Q Lu, A wire-driven drilling device (**Patent Number: CN106703720B**), China University of Petroleum Beijing, 2018, China

H Wang, G Li, **M Wang**, S Tian, Z Huang, B Yang, Q Lu, L Weng, Supercritical CO₂ sand carrying simulation experimental device and system in fractures (**Patent Number: CN108590620B**), China University of Petroleum Beijing, 2020, China

Presentations

CONTRIBUTED PRESENTATIONS

M Wang, B Freeman, L Katz, 2026. Incorporating Ion Pairing into an Extended Pitzer Model for Activity Coefficients: Implications for Ion Exchange Membrane Modeling, International Congress on Membranes and Membrane Processes (ICOM), San Antonio, TX, USA

L Katz, **M Wang**, Z Bajalan, Z Huang, J Cui, N. Vinh, B Freeman, G. Henkelman, 2026. The Role of Ion Pairing in Membrane Separations, 2026 Gordon Research Conference on Environmental Sciences: Water, Holderness, NH, USA

M Wang, Z Huang, J Cui, N. Vinh, B Freeman, L Katz, 2025. Quantifying Ion Pairing in Monovalent Sulfate Salts: Implications for Ion Exchange Membrane Modeling, Association of Environmental Engineering and Science Professors (AEESP) 2025 Research and Education Conference, Durham, NC, USA

- M Wang**, J Cui, N. Vinh, Z Huang, B Freeman, L Katz, 2025. Comprehensive Analysis of Ion Pairing and Solvation in Monovalent Sulfate Salts: Insights from Raman and Dielectric Spectroscopy. American Chemical Society Spring 2025, San Diego, CA, USA
- M Wang**, H Nguyen, S Louder, A Asatekin, D Rodrigues, 2021. Calcium Sulfate Formation on Different Zwitterionic Amphiphilic Copolymer Substrates: Implication for Developing Anti-scaling Surfaces. American Chemical Society Southwest Regional Meeting (SWRM), Austin, TX, USA
- M Wang**, B Cao, Y Hu. 2019. Effects of Organic Macromolecule on Gypsum Scaling of Reverse Osmosis Membrane. Water Quality Technology Conference, Dallas, TX, USA

Awards, Fellowships, & Grants

2025	Processes 2024 Outstanding Reviewer Award	\$500
2025	American Chemical Society GEOC Early Career Award	\$1000
2021	Graduate Student Travel Award , ACS Southwest Regional Meeting	
2018–2023	Graduate Tuition Fellowship , University of Houston	\$ 33,000/year
2017	Robert Rocca Scholarship from Tenaris China	RMB 10,000
2017	The First-class Scholarship , China University of Petroleum Beijing	RMB 96,00
2016	Outstanding Graduate Student , China University of Petroleum Beijing	
2016	Studies Excellent Scholarship , High-Pressure Water Jet Laboratory, China University of Petroleum	RMB 1000
2016	3rd Prize in the 13th National Post-Graduate Mathematic Contest in Modeling	
2015	3rd Prize in the 3th China Petroleum Engineering Design Competition	

Skills

MATERIAL CHARACTERIZATIONS

Atomic force microscopy, Vertical scanning interferometry, Attenuated total reflection Fourier transform IR, X-ray photoelectron spectroscopy, X-ray powder diffraction, Scanning electron microscopy, Transmission electron microscopy, Zetasizer Nano ZS, Raman, Grazing Incidence Small Angle X-ray Scattering (GISAXS/SAXS), and Grazing Incidence Wide-Angle X-ray Scattering (GIWAXS)

ANALYTICAL TECHNIQUES

Inductively coupled plasma mass spectrometry, Atomic Absorption Spectroscopy, Total organic carbon measurement, UV/Vis spectrometry, Ion Chromatography Mass Spectrometry, Quartz Crystal Microbalance with Dissipation (QCMD), Raman and Dielectric spectroscopy

MODELING AND SOFTWARE SKILLS

AutoCAD, ANSYS Fluent, ICEM, CFD-Post, Solid works, Machine/Deep Learning, Geochemist's Workbench (GWB)

PROGRAMMING SKILLS

C/C++, Java, Python, MATLAB, Git, PyTorch, Linux Shell, HTML/CSS, SQL, PL/SQL, MATLAB

Peer Reviewer

Conducted a total of 105 peer reviews across numerous journals, including **MDPI Energies, Geotechnics, Sustainability, Land, Applied Sciences, Water, Materials, Foundations, Fluids, Mathematics, Processes, NPJ Clean Water, and Frontiers Earth Science, Energy Science & Engineering, Water Resources Research, and Physics of Fluids.**